

Abhishek Kumar

Career Objective

To build an internationally recognized research program in machine learning and evolutionary computation by advancing representation learning, intelligent optimization algorithms, and trustworthy artificial intelligence, while contributing to high-quality research, teaching, and collaborative innovation that addresses challenging real-world problems.

Education

- 2020 **Ph.D, Systems Engineering**, *Indian Institute of Technology (BHU)*, Varanasi, India.
- 2013 **B.Tech, Electrical Engineering**, *Uttarakhand Technical University*, Dehradun, India.
Percentage: 78%

Doctoral Thesis

Title *Mathematical Algorithms for Power Distribution Systems.*

Supervisor Dr. Rakesh Kumar Misra, Dr. Devender Singh

- Contribution
- Developed mathematically grounded optimization algorithms for solving complex power system analysis and operational planning problems, including load flow and optimal power flow in modern distribution networks.
 - Reformulated challenging power system problems as mathematical optimization problems, enabling the application of evolutionary computation and nature-inspired optimization algorithms to improve robustness, convergence, and solution quality.
 - Designed efficient optimization algorithms and constraint-handling techniques capable of solving highly nonlinear, ill-conditioned, and constrained engineering optimization problems encountered in intelligent power distribution systems.
 - Proposed scalable iterative solution strategies for islanded microgrids and distributed energy systems, improving computational efficiency and numerical stability under practical operating conditions.
 - Established a unified optimization framework that integrates mathematical modeling, computational intelligence, and artificial intelligence techniques for solving complex real-world engineering problems, providing a foundation for subsequent research in machine learning, evolutionary optimization, and intelligent decision-making systems.

Work Experience

- 2024/11/21 **Visiting Professor**, *Department of Computer Science, Faculty of Electrical Engineering and Computer Science, VSB – Technical University of Ostrava, Ostrava.*
to current Czech Republic-70800
- 2024/03/01 **Assistant Professor**, *Department of Computer Science*, Sejong University, Seoul.
to Republic of Korea-05006
- 2024/10/17

2023/05/01 **BK21 Postdoctorate Fellow**, *Department of Artificial Intelligence*, Kyungpook National
to University, Daegu.
2024/02/28 Republic of Korea-41566
2022/05/24 **Postdoctorate Fellow**, *LG Research Centre, Department of Artificial Intelligence*, Kyung-
to pook National University, Daegu.
2023/05/23 Republic of Korea-41566
2021/05/24 **BK21 Postdoctorate Fellow**, *Department of Artificial Intelligence*, Kyungpook National
to University, Daegu.
2022/05/23 Republic of Korea-41566
2020/01/24 **Visiting Scholar**, *ECIS Lab, Department of Artificial Intelligence*, Kyungpook National
to University, Daegu.
2021/05/23 Republic of Korea-41566

Research Interest

My research focuses on developing principled machine learning and evolutionary optimization algorithms with strong theoretical foundations and practical applicability. My work primarily spans representation learning, deep clustering, evolutionary computation, neural architecture search, optimization theory, and their applications to computer vision and real-world optimization problems.

- **Representation Learning and Deep Clustering:** Developing robust unsupervised and self-supervised learning frameworks by integrating contrastive learning, pseudo-supervision, metric learning, and optimization techniques for high-dimensional noisy data. The goal is to learn discriminative feature representations and automatically discover meaningful cluster structures with minimal supervision.
- **Evolutionary Computation and Optimization Theory:** Designing efficient evolutionary optimization algorithms, including Differential Evolution, CMA-ES, Spherical Search, and surrogate-assisted optimization, for expensive, large-scale, constrained, and real-world optimization problems. Particular interests include adaptive search strategies, parameter adaptation, theoretical analysis, and benchmark development.
- **Machine Learning Optimization:** Developing optimization techniques for neural architecture search, model compression, channel pruning, hyperparameter optimization, and automated machine learning (AutoML), with emphasis on improving computational efficiency, scalability, and generalization.
- **Computer Vision and Intelligent Systems:** Applying modern machine learning and optimization techniques to computer vision problems, including image segmentation, object detection, scene understanding, domain adaptation, and multimodal learning.

I am also interested in the following research areas.

- Contrastive, Self-supervised, and Foundation Model Learning,
- Evolutionary Multi-objective and Many-objective Optimization,
- Constraint Handling Techniques,
- Large-scale and Expensive Optimization,
- Surrogate-assisted Optimization,
- Explainable, Trustworthy, and Robust Artificial Intelligence,
- AutoML, Neural Architecture Search, and Hyperparameter Optimization,
- Applications of Evolutionary Optimization in Real-world Engineering Problems.

Research Qualification

- ↪ SCI-indexed peer-reviewed journal (Main Author): 21
- ↪ SCI-indexed peer-reviewed journal (Co-Author): 18
- ↪ SCOPUS-indexed peer-reviewed journal (Main Author): 1
- ↪ SCOPUS-indexed peer-reviewed journal (Co-Author): 2
- ↪ Top Conferences in Computer Science and Artificial Intelligence (Main Author): 6
- ↪ Conference Publications/ Symposia/ Workshops (Main Author): 1
- ↪ Conference Publications/ Symposia/ Workshops (Co-Author): 5
- ↪ SCOPUS-indexed Book Chapters: 3
- ↪ Citations: 2743
- ↪ h-index: 23
- ↪ i10-index: 40

List of Publications

⊛ Top Conferences in Computer Science and Artificial Intelligence.

1. **Abhishek Kumar**, Swagatam Das, and Rammohan Mallipeddi. “*UEQMS: UMAP Embedded Quick Mean Shift Algorithm for High Dimensional Clustering.*” *The 37th AAAI Conference on Artificial Intelligence, 2023, Accepted for Oral Presentation, (AAAI’23) (A* Conference).*
2. **Abhishek Kumar**, Oladayo S. Ajani, Swagatam Das, and Rammohan Mallipeddi. “*Grid-shift: A Faster Mode-seeking Algorithm for Image Segmentation and Object Tracking.*” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, Accepted for Oral Presentation, (CVPR’22) (A* Conference, Ranked first and fourth as Top publication in Computer Science and all category, respectively according to Google Scholar).*
3. **Abhishek Kumar**, Swagatam Das, and Ivan Zelinka. “*A self-adaptive spherical search algorithm for real-world constrained optimization problems.*” *Proceedings of the 2020 Genetic and Evolutionary Computation Conference Companion, 2020, (GECCO’20) (A Conference, Winner of the Special Session & Competitions on Real-World Single Objective Constrained Optimization jointly organized by GECCO’20 and CEC’20).*
4. **Abhishek Kumar**, Swagatam Das, and Ivan Zelinka. “*A modified covariance matrix adaptation evolution strategy for real-world constrained optimization problems.*” *Proceedings of the 2020 Genetic and Evolutionary Computation Conference Companion, 2020, (GECCO’20) (A Conference, Ranked third in the Special Session & Competitions on Real-World Single Objective Constrained Optimization jointly organized by GECCO’20 and CEC’20).*
5. **Abhishek Kumar**, Rakesh Kumar Misra, Devender Singh and Swagatam Das. “*Testing A Multi-Operator based Differential Evolution Algorithm on the 100-Digit Challenge for Single Objective Numerical Optimization.*” *2019 IEEE Congress on Evolutionary Computation (CEC). IEEE, 2019, (CEC’19) (B Conference, Ranked Fifth in the Special Session & Competition on 100-Digit Challenge and Numerical Optimization Scenarios jointly organized by GECCO’19 and CEC’19).*

6. **Abhishek Kumar**, Rakesh Kumar Misra, and Devender Singh “*Improving the local search capability of effective butterfly optimizer using covariance matrix adapted retreat phase.*” *2017 IEEE Congress on Evolutionary Computation (CEC), IEEE, 2017*, (CEC’17) (**B Conference, Winner of the Special Session & Competitions on Real-Parameter Single Objective Optimization organized by CEC’17**).

⊛ Refereed International Journal Articles.

1. **Abhishek Kumar**, and Dong-Gyu Lee “*Addressing class collision and representation collapse in contrastive clustering via Pseudo-supervised triplet and InfoNCE losses.*” *Applied Soft Computing* (2026): 115751. (ISSN:1568-4946), (SCI Expanded, Impact Factor: 7.800) (**WOS: Q1(86.8%) in Computer Science, Interdisciplinary Applications**).
2. **Abhishek Kumar**, Athul Shibu, and Dong-Gyu Lee “*Automatic Network Compression via Evolutionary Search-guided Channel Pruning of Deep Convolutional Neural Networks*” *Applied Soft Computing* (2025): 113521. (ISSN:1568-4946), (SCI Expanded, Impact Factor: 7.800) (**WOS: Q1(86.8%) in Computer Science, Interdisciplinary Applications**).
3. **Abhishek Kumar**, Swagatam Das, and Václav Snášel. “*Efficient Three-Stage Surrogate-Assisted Differential Evolution for Expensive Optimization Problems.*” *Swarm and Evolutionary Computation* 98 (2025): 102093. (ISSN: 2210-6502), (SCI Expanded, Impact Factor: 9.600) (**WOS: Q1(94.2%) in Computer Science, Artificial Intelligence**).
4. **Abhishek Kumar**, Oladayo S Ajani, Swagatam Das, and Rammohan Mallipeddi “*Entropy-weighted medoid shift: An automated clustering algorithm for high-dimensional data.*” *Applied Soft Computing* 169 (2025): 112347. (ISSN:1568-4946), (SCI Expanded, Impact Factor: 7.800) (**WOS: Q1(86.8%) in Computer Science, Interdisciplinary Applications**).
5. Hye-Seong Hong, Jeong-Cheol Lee, **Abhishek Kumar**, Sangtae Ahn, Dong-Gyu Lee. “*Deep scene understanding with extended text description for human object interaction detection.*” *Expert Systems with Applications* 259 (2025): 125297. (ISSN: 0957-4174), (SCI Expanded, Impact Factor: 9.400) (**WOS: Q1(95.0%) in Engineering, Electronics and Electric**).
6. **Abhishek Kumar**, and Dong-Gyu Lee. “*Enhancing clustering representations with positive proximity and cluster dispersion learning.*” *Information Sciences* 686 (2025): 121277. (ISSN: 1872-6291), (SCI Expanded, Impact Factor: 6.800) (**WOS: Q1(92.1%) in Computer Science, Information Systems**).
7. Oladayo S Ajani, **Abhishek Kumar**, and Rammohan Mallipeddi “*Covariance matrix adaptation evolution strategy based on correlated evolution paths with application to reinforcement learning.*” *Expert Systems with Applications* 246 (2024): 123289. (ISSN: 0957-4174), (SCI Expanded, Impact Factor: 9.400) (**WOS: Q1(95.0%) in Engineering, Electronics and Electric**).
8. **Abhishek Kumar**, Anish Chakrabarty, Sankha Subhra Mullick, and Swagatam Das “*Enhancing Contrastive Clustering with Negative Pair-guided Regularization.*” *Transactions on Machine Learning Research*, (SCOPUS Indexed).

9. Abhimanyu Kumar, **Abhishek Kumar**, Rammohan Mallipeddi, and Dong-Gyu Lee. “**High-Density Cluster Core-based k -Means Clustering with an Unknown Number of Cluster.**” *Applied Soft Computing*, **155**, 111419. (ISSN: 1568-4946), (SCI Expanded, Impact Factor: 7.800) (**WOS: Q1(86.8%) in Computer Science, Interdisciplinary Applications**).
10. **Abhishek Kumar**, Swagatam Das, and Rammohan Mallipeddi. “**An Efficient Differential Grouping Algorithm for Large-Scale Global Optimization.**” *IEEE Transactions on Evolutionary Computation*, vol. **28**, no. **1**, pp. **32-46**, Feb. **2024**. (ISSN: 1089-778X), (SCI Expanded, Impact Factor: 15.900) (**WOS: Q1(96.3%) in Computer Science, Theory and Methods**).
11. Hye-Seong Hong, **Abhishek Kumar**, and Dong-Gyu Lee. “**Robust unsupervised domain adaptation by retaining confident entropy via edge concatenation.**” *Expert Systems with Applications* **238** (2024): **122120**. (ISSN: 0957-4174), (SCI Expanded, Impact Factor: 9.400) (**WOS: Q1(95.0%) in Operations Research & Management Science**).
12. Satyawant Kumar, **Abhishek Kumar**, Dong-Gyu Lee. “**RSSGLT: Remote Sensing Image Segmentation Network based on Global-Local Transformer.**” *IEEE Geoscience and Remote Sensing Letters*, vol. **21**, pp. **1-5**, **2024**. (ISSN: 1545-598X), (SCI Expanded, Impact Factor: 4.800) (**WOS: Q1(90.8%) in Geochemistry & Geophysics**).
13. **Abhishek Kumar**, Swagatam Das, Lingping Kong and Vaclav Snasel “**Spherical Search with A Low Precision Projection Matrix for Real-World Optimization.**” *IEEE Transactions on Cybernetics*, vol. **53**, no. **7**, pp. **4107-4121**, **July 2023**. (ISSN:2168-2267), (SCI Expanded, Impact Factor: 11.300) (**WOS: Q1(98.5%) in Computer Science, Cybernetics**).
14. **Abhishek Kumar**, Bablesh Kumar Jha, Swagatam Das, and Rammohan Mallipeddi. “**Spherical Search based Constrained Optimization Algorithm for Power Flow Analysis of Islanded Microgrids.**” *Applied Soft Computing*, **136** (2023): **110057**. (ISSN: 1568-4946), (SCI Expanded, Impact Factor: 7.800) (**WOS: Q1(86.8%) in Computer Science, Interdisciplinary Applications**).
15. Athul Shibu, **Abhishek Kumar**, Heechul Jung and Dong-Gyu Lee. “**Rewarded Meta-pruning: Meta Learning with Rewards for Channel Pruning**” *Mathematics*, **2023; 11(23):4849**. (ISSN: 2227-7390), (SCI Expanded, Impact Factor: 2.300) (**WOS: Q1(94.4%) in Mathematics**).
16. Yang Zhuji, **Abhishek Kumar**, Rammohan Mallipeddi, and Dong-Gyu Lee. “**Spherical search with epsilon constraint and gradient-based repair framework for constrained optimization.**” *Swarm and Evolutionary Computation* **82** (2023): **101370**. (ISSN: 2210-6502), (SCI Expanded, Impact Factor: 9.600) (**WOS: Q1(94.2%) in Computer Science, Theory and Methods**).
17. Bishal Karmarkar, **Abhishek Kumar**, and Rammohan Mallipeddi. “**CMA-ES with Exponential based Multiplicative Covariance Matrix Adaptation for Global Optimization.**” *Swarm and Evolutionary Computation*, **79** (2023): **101296**. (ISSN: 2210-6502), (SCI Expanded, Impact Factor: 9.600) (**WOS: Q1(94.2%) in Computer Science, Artificial Intelligence**).
18. Kenneth V. Price, **Abhishek Kumar**, and P. N. Suganthan. “**Trial-Based Dominance for Comparing both the Speed and Accuracy of Stochastic Optimizers with Standard Non-Parametric Tests.**” *Swarm and Evolutionary Computation*, **78** (2023): **101287**. (ISSN: 2210-6502), (SCI Expanded, Impact Factor: 9.600) (**WOS: Q1(94.2%) in Computer Science, Artificial Intelligence**).

19. **Abhishek Kumar**, Swagatam Das, Rakesh Kumar Misra, and Devender Singh. *“A v-constrained Matrix Adaptation Evolution Strategy with Broyden-based Mutation for Constrained Optimization.”* *IEEE Transactions on Cybernetics*, vol. 52, no. 6, pp. 4784-4796, June 2022. (ISSN:2168-2267), (SCI Expanded, Impact Factor: 11.300) (**WOS: Q1(98.5%) in Computer Science, Cybernetics**).
20. **Abhishek Kumar**, Swagatam Das and Rammohan Mallipeddi. *“A Reference Vector based Simplified Covariance Matrix Adaptation Evolution Strategy for Constrained Global Optimization.”* *IEEE Transactions on Cybernetics*, vol. 52, no. 5, pp. 3696-3709, May 2022. (ISSN:2168-2267), (SCI Expanded, Impact Factor: 11.300) (**WOS: Q1(98.5%) in Computer Science, Cybernetics**).
21. **Abhishek Kumar**, Partha Biswas, and Ponnuthurai Suganthan. *“Differential Evolution with Orthogonal Array-based Initialization and a Novel Selection Strategy.”* *Swarm and Evolutionary Computation* 68 (2022): 101010. (ISSN: 2210-6502), (SCI Expanded, Impact Factor: 9.600) (**WOS: Q1(94.2%) in Computer Science, Artificial Intelligence**).
22. Satyawant Kumar, **Abhishek Kumar**, Dong-Gyu Lee. *“Semantic Segmentation of UAV Images Based on Transformer Framework with Context Information.”* *Mathematics*, 2022; 10(24):4735. (ISSN: 2227-7390), (SCI Expanded, Impact Factor: 2.3) (**WOS: Q1(94.4%) in Mathematics**).
23. Amit Kumar, Basanth Kumar Sethi, **Abhishek Kumar**, Devender Singh, and Rakesh Kumar Misra. *“Three-Level Hierarchical Management of Active Distribution System With Multimicrogrid.”* *IEEE Systems Journal*, 17.1 (2022): 605-616. (ISSN: 1932-8184), (SCI Expanded, Impact Factor: 4.900) (**WOS: Q1(80.3%) in Operations Research and Management Science**).
24. **Abhishek Kumar**, Swagatam Das, and Václav Snášel. *“Improved Spherical Search with Local Distribution induced Self-Adaptation for Hard Non-convex Optimization with and without Constraints.”* *Information Sciences*, 615 (2022): 604-637. (ISSN: 1872-6291), (SCI Expanded, Impact Factor: 6.800) (**WOS: Q1(92.1%) in Computer Science, Information Systems**).
25. Abhimanyu Kumar, **Abhishek Kumar**, Rammohan Mallipeddi, and Dong-Gyu Lee. *“Adaptive Backward/Forward Sweep for Solving Power Flow of Islanded Microgrids.”* *Energies*, 15.24 (2022): 9348. (ISSN: 1996-1073), (SCI Expanded, Impact Factor: 3.900) (**WOS: Q3(37.4%) in Energy & Fuels**).
26. Oladayo S Ajani, **Abhishek Kumar**, Rammohan Mallipeddi, Swagatam Das, Ponnuthurai Nagaratnam Suganthan. *“Benchmarking Optimization-Based Energy Disaggregation Algorithms.”* *Energies* 15.5 (2022): 1600. (ISSN:1996-1073), (SCI Expanded, Impact Factor: 3.700) (**WOS: Q3(37.4%) in ENERGY & FUELS**).
27. **Abhishek Kumar**, Swagatam Das, and Rammohan Mallipeddi. *“An Inversion-free Robust Power Flow Algorithm for Microgrids.”* *IEEE Transactions on Smart Grid* 12.4 (2021). (ISSN:1949-3053), (SCI Expanded, Impact Factor: 10.1), (**WOS: Q1(95.0%) in Electrical & Electronic Engineering**).
28. **Abhishek Kumar**, Guohua Wo, Mostafa Ali, Qizhang Luo, Rammohan Mallipeddi, Ponnuthurai Suganthan and Swagatam Das. *“A Benchmark-Suite of Real-World Constrained Multi-Objective Optimization Problems and some Baseline Results.”* *Swarm and Evolutionary Computation* 67 (2021): 100961. (ISSN: 2210-6502), (SCI Expanded, Impact Factor: 9.600) (**WOS: Q1(94.2%) in Computer Science, Artificial Intelligence**).

29. **Abhishek Kumar**, Bablesh Kumar Jha, Swagatam Das, and Rammohan Mallipeddi. *“Power Flow Analysis of Islanded Microgrids: A Differential Evolution Approach” IEEE Access 9 (2021): 61721-61738.* (ISSN:2169-3536), (SCI Expanded, Impact Factor: 4.2) (**WOS: Q2(66.5%) in Electrical & Electronic Engineering**).
30. Bablesh Kumar Jha, Amit Singh, **Abhishek Kumar**, Devender Singh and Rakesh Kumar Misra. *“Phase Unbalance and PAR Constrained Optimal Active and Reactive Power Scheduling of Virtual Power Plants (VPPs).” International Journal of Electrical Power & Energy Systems (2021) 125, 106443.* (ISSN:0142-0615), (SCI Expanded, Impact Factor: 5.3) (**WOS: Q1(77.9%) in Electrical & Electronic Engineering**).
31. **Abhishek Kumar**, Guohua Wo, Mostafa Ali, Rammohan Mallipeddi, Ponnuthurai Suganthan and Swagatam Das. *“A Test-suite of Non-Convex Constrained Optimization Problems from the Real-World and Some Baseline Results.” Swarm and Evolutionary Computation, vol. 56, Aug. 2020.* (ISSN: 2210-6502), (SCI Expanded, Impact Factor: 9.600) (**WOS: Q1(94.2%) in Computer Science, Artificial Intelligence**).
32. **Abhishek Kumar**, Bablesh Kumar Jha, Dharmender Kumar Dheer, Devender Singh, and Rakesh Kumar Misra. *“A Nested-Iterative Newton-Raphson based Power Flow Formulation for Droop-based Islanded Microgrids” Electric Power Systems Research, vol.180, March 2020.* (ISSN:0378-7796), (SCI Expanded, Impact Factor: 4.3) (**WOS: Q2(67.3%) in Electrical & Electronic Engineering**).
33. **Abhishek Kumar**, Bablesh Kumar Jha, Rakesh Kumar Misra, and Devender Singh. *“A New Current Injection Based Power flow Formulation.” Electric Power Components and Systems 48.3 (2020): 268-280.* (ISSN: 1532-5016), (SCI Expanded, Impact Factor: 1.9) (**WOS: Q3(31.0%) in Electrical & Electronic Engineering**).
34. Bablesh Kumar Jha, **Abhishek Kumar**, Dharmendra Kumar Dheer, Devender Singh and Rakesh Kumar Misra. *“A modied current injection load flow method under different load model of EV for distribution system.” International Transactions on Electrical Energy Systems, vol. 30, April. 2020.* (ISSN:2050-7038), (SCI Expanded, Impact Factor: 2.6) (**WOS: Q3(47.6%) in Electrical & Electronic Engineering**).
35. Bablesh Kumar Jha, Amit Singh, **Abhishek Kumar**, Dharmendra Kumar Dheer, Devender Singh and Rakesh Kumar Misra. *“Day ahead scheduling of PHEVs and D-BESSs in presence of DGs in distribution system.” IET Electrical Systems in Transportation 10.2 (2020): 170-184.* (ISSN:2042-9738), (SCI Expanded, Impact Factor: 2.1) (**WOS: Q3(36.4%) in Electrical & Electronic Engineering**).
36. **Abhishek Kumar**, Rakesh Kumar Misra, Devender Singh, Sujeet Mishra and Swagatam Das. *“The spherical search algorithm for bound-constrained global optimization problems.” Applied Soft Computing (2019): 105734.* (ISSN: 1568-4946), (SCI Expanded, Impact Factor: 7.800) (**WOS: Q1(86.8%) in Computer Science, Interdisciplinary Applications**).
37. **Abhishek Kumar**, Bablesh Kumar Jha, Rakesh Kumar Misra, and Devender Singh. *“Current Injection based Newton-Raphson Power flow algorithm for droop-based Islanded Microgrids.” IET Generation, Transmission, & Distribution, vol.13, Dec. 2019.* (ISSN:1350-2360), (SCI, Impact Factor: 2.8) (**WOS: Q2(51.4%) in Electrical & Electronic Engineering**).
38. **Abhishek Kumar**, Bablesh Kumar Jha, Dharmender Kumar Dheer, Devender Singh and Rakesh Kumar Misra. *“Nested backward/forward sweep algorithm for power flow analysis of droop regulated islanded microgrids.” IET Generation, Transmission & Distribution 13.14 (2019): 3086-3095.* (ISSN:1350-2360), (SCI, Impact Factor: 2.8) (**WOS: Q2(51.4) in Electrical & Electronic Engineering**).

39. Tarun Maini, **Abhishek Kumar**, Rakesh Kumar Misra, and Devender Singh. *“Intelligent Fuzzy Rough Set Based Feature Selection using Swarm algorithms with improved initialization.” Journal of Intelligent and Fuzzy Systems (JIFS), vol. 27, July. 2019. (ISSN: 1875-8967), (SCI Expanded, Impact Factor: 1.0) (WOS: Q4(14.0%) in Computer Science, Artificial Intelligence).*
40. Bablesh Kumar Jha, **Abhishek Kumar**, Devender Singh, and Rakesh Kumar Misra. *“Co-ordinated effect of PHEVs with DGs on distribution network.” International Transactions on Electrical Energy Systems, vol. 29, April. 2019. (ISSN:2050-7038), (SCI Expanded, Impact Factor: 2.6) (WOS: Q3(47.6%) in Electrical & Electronic Engineering).*
41. Sujeet Mishra, **Abhishek Kumar**, Rakesh Kumar Misra, and Devender Singh. *“Feeder Phase Balancing using Single Phase Distributed Generations Considering Voltage Dependency of Loads.” Journal of Advanced Research in Dynamical and Control Systems, 2019. (ISSN: 1943-023X), (SCOPUS Indexed).*
42. Tarun Maini, Rakesh Kumar Misra, Devender Singh, and **Abhishek Kumar**. *“Rough Set Based Feature Selection Using Swarm Algorithms with Improved Initialization.” Journal of Computational and Theoretical Nanoscience 15, no. 6-7 (2018): 2350-2354. (ISSN 1546-1963), (SCOPUS Indexed).*

➤ Summary of Published Articles.

JIF QUARTILE	JIF PERCENTILE	Main-author	Co-author	Total
Q1	95%≤	5	3	8
	90-94.99%	6	6	12
	85-89.99%	5	1	6
	75-84.99%	0	2	2
	Total	16	12	28
Q2	50-74.99%	4	0	4
Q3	25-49.99%	1	5	6
Q4	0-24.99%	0	1	1
SCOPUS	—	1	2	3
Total		22	20	42

⊛ Scopus Indexed Book Chapters.

1. **Abhishek Kumar**, Tarun Maini, Rakesh Kumar Misra, and Devender Singh. *“Butterfly Constrained Optimizer for Constrained Optimization Problems.” In Computational Intelligence: Theories, Applications and Future Directions- Volume II, pp. 477-486. Advances in Intelligent Systems and Computing, vol 799.Springer, Singapore, 2019.(ISSN:978-981-13-1135-2).*
2. Sujeet Mishra, **Abhishek Kumar**, Devender Singh, and Rakesh Kumar Misra. *“Butterfly Optimizer for Placement and Sizing of Distributed Generation for Feeder Phase Balancing.” In Computational Intelligence: Theories, Applications and Future Directions- Volume II, pp. 519-530. Advances in Intelligent Systems and Computing, vol 799.Springer, Singapore, 2019.(ISSN:978-981-13-1135-2).*

3. Tarun Maini, **Abhishek Kumar**, Rakesh Kumar Misra, and Devender Singh. **“Fuzzy Rough Set-Based Feature Selection with Improved Seed Population in PSO and IDS.”** In *Computational Intelligence: Theories, Applications and Future Directions- Volume II*, pp. 137-149. *Advances in Intelligent Systems and Computing*, vol 799. Springer, Singapore, 2019. (ISSN:978-981-13-1135-2).

⊛ Refereed International Conferences/Symposiums/Workshops.

1. Satyawant Kumar, **Abhishek Kumar**, Hye-Seong Hong, and Dong-Gyu Lee. **“Encoder-Decoder based Segmentation Model for UAV Street Scene Images.”** In *2023 IEEE International Conference on Consumer Electronics (ICCE)*, pp. 1-4. IEEE, 2023.
2. Rakesh Kumar Misra, Devender Singh, and, **Abhishek Kumar** **“Spherical search algorithm: A metaheuristic for bound-constrained optimization.”** In *Optimization, Variational Analysis and Applications: IFSOVAA-2020, Varanasi, India, February 2–4*, pp. 421-441. Springer Singapore, 2021.
3. Lingping Kong, **Abhishek Kumar**, Václav Snášel, Swagatam Das, Pavel Krömer, and Varun Ojha. **“CSS–A cheap-surrogate-based selection operator for multi-objective optimization.”** In *International Conference on Bioinspired Optimization Methods and Their Applications*, pp. 54-68. Cham: Springer International Publishing, 2022.
4. Tarun Maini, **Abhishek Kumar**, Rakesh Kumar Misra, and Devender Singh. **“Rough set based feature selection using swarm intelligence with distributed sampled initialization.”** In *2017 6th International Conference on Computer Applications In Electrical Engineering-Recent Advances (CERA)*, pp. 92-97. IEEE, 2017.
5. Tarun Maini, **Abhishek Kumar**, Rakesh Kumar Misra, and Devender Singh. **“Feature selection with intelligent dynamic swarm and fuzzy rough set.”** *2017 International Conference on Computing, Communication and Automation (ICCCA)*, Greater Noida, 2017, pp. 384-388.
6. **Abhishek Kumar**, Rakesh Kumar Misra, and Devender Singh. **“Butterfly Optimizer.”** In *2015 IEEE Workshop on Computational Intelligence: Theories, Applications and Future Directions (WCI)*, pp. 1-6. IEEE, 2015.

Ongoing Research Work

- **For unsupervised and contrastive representation learning**, developing robust deep clustering frameworks by integrating contrastive learning, pseudo-supervision, metric learning, and representation regularization for high-dimensional noisy datasets.
- **For automatic clustering**, designing optimization-driven clustering algorithms capable of estimating the number of clusters automatically while improving robustness, scalability, and clustering quality.
- **For neural architecture search and model compression**, developing evolutionary optimization algorithms for neural architecture search, channel pruning, and lightweight deep neural networks with high accuracy and computational efficiency.
- **For hyper-parameter optimization**, developing efficient evolutionary optimization techniques for automatic hyper-parameter tuning of deep neural networks and machine learning models.

- ↪ **For evolutionary computation**, designing efficient differential evolution, CMA-ES, surrogate-assisted optimization, and spherical search algorithms for expensive, large-scale, and real-world optimization problems.
- ↪ **For constrained optimization**, developing advanced constraint-handling mechanisms and benchmark problem suites for solving complex single-, multi-, and many-objective constrained optimization problems.
- ↪ **For many-objective optimization**, developing adaptive search strategies, diversity preservation techniques, and selection mechanisms for irregular and high-dimensional Pareto fronts.
- ↪ **For explainable and trustworthy machine learning**, investigating robust representation learning and optimization techniques that improve the reliability, interpretability, and generalization of deep learning models.

Computer Skill

- ↪ Package: C/C++, Java, MATLAB, Julia, Python, PSCAD, ETAP, Origin, GAMS and OpenDss.
- ↪ Operating System: Windows, UNIX(Linux) and Android.
- ↪ Word Processor: MS word, MS powerpoint, LATEX, Beamer, and Overleaf.

Achievements

- ↪ The paper titled: “**The spherical search algorithm for bound-constrained global optimization problems**”, has been selected for the “**2022 Best Paper Award**” in the category “**Editor’s Selection**” in **Applied Soft Computing Journal**.(Link for more information.)
- ↪ Stood **First** for proposing optimization algorithm titled: “**SASS**”, in Special Session and Competition on Real-World Constrained Single Objective Optimization at **IEEE CEC-2020**, Glasgow, United Kingdom, 19-24 July 2020 and **GECCO-2020**, Cancun, Mexico, 8-12 July 2020.
- ↪ Stood **Third** for proposing optimization algorithm titled: “**sCMAgES**”, in Special Session and Competition on Real-World Constrained Single Objective Optimization at **IEEE CEC-2020**, Glasgow, United Kingdom, 19-24 July 2020 and **GECCO-2020**, Cancun, Mexico, 8-12 July 2020.
- ↪ Stood **Fifth** for proposing optimization algorithm titled: “**ESHADE_USM**”, in 100-Digit Challenge for Single Objective Numerical Optimization at **IEEE CEC-2019**, Wellington, New Zealand, 10-13 June 2019.
- ↪ Stood **Second** for proposing optimization algorithm titled: “**EBOWithCMAR**”, in Non-tuning Algorithm Section in 100-Digit Challenge for Single Objective Numerical Optimization at **IEEE CEC-2019**, Wellington, New Zealand, 10-13 June 2019.
- ↪ Stood **First** for presenting poster/model titled: “**Effective Butterfly optimizer**”, in the Institute Day at Indian Institute of Technology (BHU), Varanasi, 16-18 Feb 2018.
- ↪ Stood **First** for proposing optimization algorithm titled: “**Effective Butterfly Optimizer (EBOWithCMAR)**”, in Special Session and Competition on Real- Parameter Single Objective Optimization at **IEEE CEC-2017**, Donostia, Spain, 5-8 June 2017.
- ↪ Recipient of “**Young Researcher Awards-2016**” from “CIS Chapter, IEEE UP Section”, IIT Kanpur, Kanpur.
- ↪ Recipient of scholarship for 5 Years from MHRD, India for Ph.D through GATE examination.
- ↪ Remained in **Top 3** throughout my **B.Tech** course.

Professional Activity

A. Editorial Activities.

- **Associate Editor** since 2025 in the journal *Computers and Electrical Engineering*. (SCI Expanded, IF: 5.500) (**WOS**: Q1(10/60) in **Computer Science, Hardware and Architecture**)
- **Associate Editor** from 2023-25 in the journal *Swarm and Evolutionary Computation*. (SCI Expanded, IF: 9.600) (**WOS**: Q1(13/189) in **Computer Science, Artificial Intelligence**)
- **Special Issue Editor** in the journal *Energies* for the Special Issue “Computational Methods and Artificial Intelligence Studies in Smart Grids”.(SCI Expanded, IF: 3.252) (**WOS**: Q3(93/145) in **ENERGY & FUELS**)
- **Special Issue Main Editor** in the journal *Swarm and Evolutionary Computation* for the Special Issue “Recent Advances in Constraint Handling Techniques for Solving Constrained Optimization Problems”. (SCI Expanded, IF: 9.600) (**WOS**: Q1(13/189) in **Computer Science, Artificial Intelligence**)

B. Regular Reviewer of the following international journals.

- **IEEE Transactions on Evolutionary Computation**
- **IEEE Transactions on Cybernetics**
- **IEEE Transactions on Systems, Man, and Cybernetics, Part A**
- **IEEE Transactions on Smart Grid**
- **Engineering Applications of Artificial Intelligence, Elsevier**
- **Information Sciences, Elsevier**
- **Swarm and Evolutionary Computations, Elsevier**
- **Applied Soft Computing, Elsevier**
- **Artificial Intelligence Review, Springer**

C. Organizing Session in International Conferences.

- Serving as a Co-organizer in Special Session & Competition on Single Objective Bound Constrained Optimization GECCO 2022, Boston, USA, 9-13 July 2022.
- Serving as a Co-organizer in Special Session & Competition on Single Objective Bound Constrained Optimization WCCI-CEC-2022, Padova, Italy, 18-23 July 2022.
- Served as a Co-organizer in Special Session & Competitions on Real-World Multi- Objective Constrained Optimization at IEEE CEC-2021, Krakow, Poland, 28 June -01 July 2021.
- Served as a Co-organizer in Special Session & Competitions on Single Objective Bound Constrained Optimization at IEEE CEC-2021, Krakow, Poland, 28 June -01 July 2021.
- Served as a Co-organizer in Special Session & Competitions on Real-World Multi-Objective Constrained Optimization at GECCO 2021, Lille, France, 10-14 July 2021.
- Served as a Co-organizer in Special Session & Competitions on Single Objective Bound Constrained Optimization at GECCO 2021, Lille, France, 10-14 July 2021.
- Served as a Co-organizer in Special Session & Competitions on Real-World Single Objective Constrained Optimization at IEEE CEC-2020, Glasgow, UK, 19-24 July 2020.
- Served as a Co-organizer in Special Session & Competitions on Real-World Single Objective Constrained Optimization at SEMCCO 2020, Wuhan, China, 12-14 June 2020.
- Served as a Co-organizer in Special Session & Competitions on Real-World Single Objective Constrained Optimization at GECCO 2020, Cancun, Mexico, 8-12 July 2020.
- Key-organising Committee Member in IEEE UPCON 2016 Conference at Indian Institute of Technology (BHU) Varanasi, Varanasi, India.

D. Teaching Experience.

Served as a tutor for undergraduate and graduate courses in Artificial Intelligence, Computer Science, and Electrical Engineering, including:

- Artificial Intelligence
- Computer Networks
- Operations Research
- Computer Methods in Power Systems
- Digital Techniques and Instrumentation
- Network Theory

E. Membership.

- ↪ Serving as a Convener, CIS Professional Activities (Executive Committee Member) in Computational Intelligence Society Chapter, IEEE UP Section (IEEECIS11), Indian Institute of Technology Kanpur (January 2022 - December 2022).
- ↪ Served as a Senior Advisor in IEEE Student Chapter, Indian Institute of Technology (BHU) Varanasi (March 2018 - March 2020).
- ↪ Served as a Vice Convener, CIS workshop/ conference Activities (Executive Committee Member) in Computational Intelligence Society Chapter, IEEE UP Section (IEEECIS11), Indian Institute of Technology Kanpur (January 2019 - December 2019).
- ↪ Student Membership: IEEE Computational Intelligence Society.

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